Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B. TECH (BIOTECHNOLOGY)
END SEMESTER EXAMINATIONS (REGULAR), NOV/DEC 2019
SUBJECT: FOOD PROCESSING TECHNOLOGY [BIO 4016]
REVISED CREDIT SYSTEM

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitable assumed.

1A.	Explain the concept of a_w and its relevance in food preservation. What are the methods adopted to control the a_w in a food?	**4**
1B.	What are the main causes of quality deterioration and spoilage of foods?	**3**
1C.	Discuss the role of minerals in human metabolism.	**3**
2A.	What are the different types of proteins present in milk? How are they used in different food products?	**3**
2B.	What are non-sugar sweeteners (NSS) used in food industry. Give example for NSS.	**2**
2C.	Diagrammatically represent the different factors influencing the microbial activity. Explain each factor in detail.	**3**
2D.	Distinguish between food infections and food intoxications.	**2**
3A.	Illustrate the hierarchical classification of food preservation techniques	**2**
3B.	What are the advantages of low-temperature processing and preservation of foods?	**2**
3C.	What factors cause spoilage of foods during storage? How is spoilage of food prevented during storage?	**2**
3D.	A food product is to be air cooled immediately after a pasteurization process. This can be done by using either natural or forced convection. Which one will be the best method and justify?	**2**
3E.	Blanching of fresh produce is aimed at reducing or eliminating enzymatic activity. Briefly name and discuss enzymes that can affect the quality of un-blanched fresh produce.	**2**

4A.	Discuss the steps involved in milk powder production, with special attention to heat treatment, evaporation and homogenization	**4**
4B.	Enlist the unit operations in processing of fruits? What are the various products from a fruit processing unit?	**4**
4C.	Explain the application of biotechnology in food product formulation and in food industry.	**2**
5A.	Explain the steps involved in HACCP & QACCP based food safety system.	**3**
5B.	Draw a simplified process block diagram for milk chocolate, chocolate syrup, and cocoa powder and explain each operation.	**4**
5C.	Food is milled from 6mm to 0.0012 mm using a 10hp motor. Would this motor be adequate to reduce the size of the particles to 0.0008mm? Assume Rittinger and kick law and that 1hp=745.7W.	**3**